

SLIDE 01 • CLIMATE SUMMARY

COORDINATES: 22.60°N, 89.50°E | DATA PERIOD: 2014 – 2024

EXECUTIVE METRICS

4.92

SOLAR (KWH/M²/D)

25.3°C

MEAN TEMP

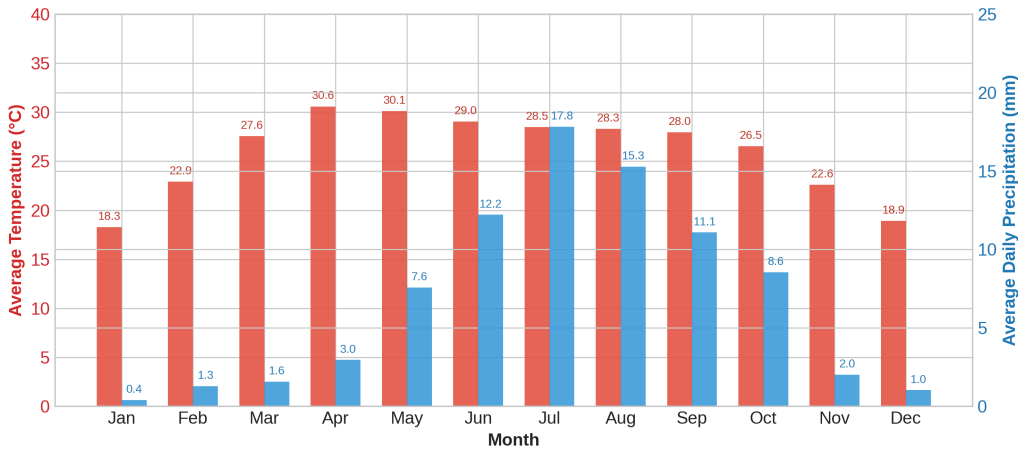
30.6°C

MAX PEAK TEMP

17.8 mm

PEAK DAILY RAIN

Monthly Climate Summary (2014 - 2024)
Latitude: 22.6°N, Longitude: 89.5°E



TEMPERATURE PROFILE

- **Peak Temp:** April (30.6°C mean) & May (30.1°C mean).
- **Lowest Temp:** January (18.3°C mean).

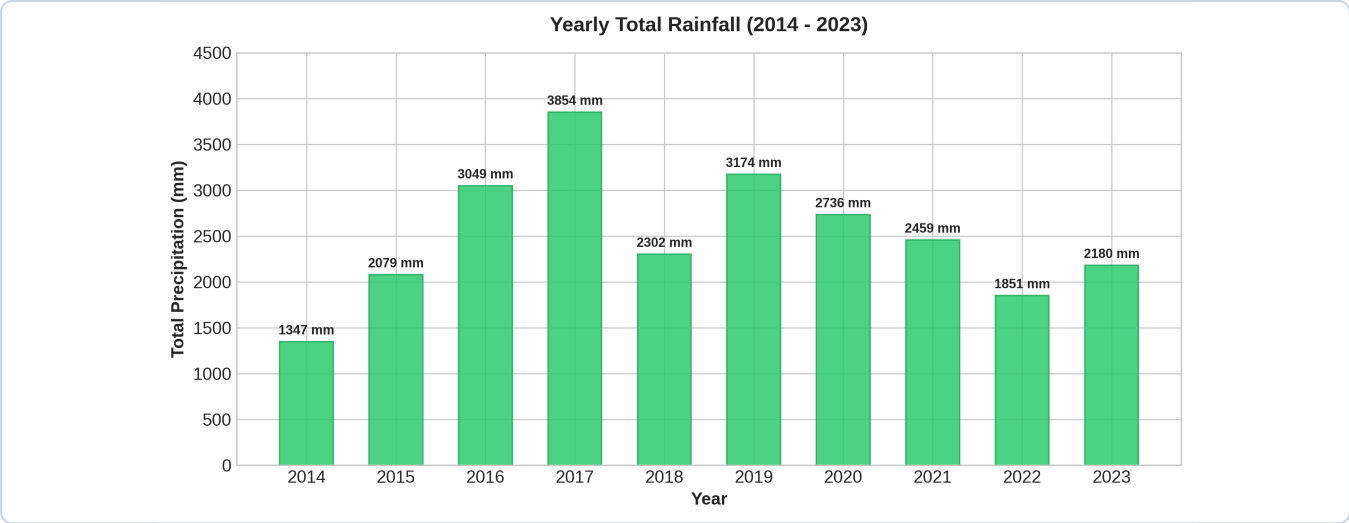
PRECIPITATION PROFILE

- **Monsoon Peak:** July (17.8 mm/day avg).
- **Dry Period:** Dec–Jan (< 0.5 mm/day avg).

SLIDE 02 • ANNUAL PRECIPITATION TREND

10-YEAR HISTORICAL RAINFALL DATA (2014 – 2023)

YEARLY ANALYSIS



EXACT MONTHLY AVERAGES MATRIX

MONTH	SOLAR (kWh/m²/d)	AVG TEMP (°C)	PRECIPITATION (mm/d)	STATUS
Jan	4.35	18.3	0.4	Dry / Cool
Feb	5.12	22.9	1.3	Dry / Moderate
Mar	5.88	27.6	1.6	Moderate
Apr	6.45 (Max)	30.6 (Max)	3.0	Peak Solar / Hot
May	6.10	30.1	7.6	Pre-Monsoon
Jun	4.75	29.0	12.2	Monsoon
Jul	4.15	28.5	17.8 (Max)	Heavy Monsoon
Aug	4.25	28.3	15.3	Monsoon
Sep	4.30	28.0	11.1	Monsoon
Oct	4.85	26.5	8.6	Post-Monsoon
Nov	4.60	22.6	2.0	Dry / Moderate
Dec	4.20	18.9	1.0	Dry / Cool

SLIDE 03 • SOLAR & ENGINEERING SPECIFICATIONS

IRRADIANCE STABILITY & DESIGN PARAMETERS

TECHNICAL DECK

SOLAR IRRADIANCE STABILITY

- **10-Year Range:** 4.86 – 4.98 kWh/m²/day.
- **Annual Variance:** < 2.5% (Extremely stable).
- **Best Generation Window:** Feb – May.

RAINFALL EXTREMES

- **Max Annual Rain:** 3,854 mm (Year 2017).
- **Min Annual Rain:** 1,347 mm (Year 2014).
- **10-Year Average:** 2,423 mm/year.

ENGINEERING & SYSTEM DESIGN DIRECTIVES

TILT ANGLE

22.6° SOUTH

Fixed angle aligned to latitude for optimal year-round yield.

TEMP DERATING

-0.35% / °C

Required coefficient for panels due to summer heat >35°C.

INCLOSURE RATING

IP67 RATED

Mandatory protection against heavy monsoon rainfall (July peak).

SUMMARY CONCLUSION

The 22.60°N, 89.50°E site demonstrates high solar viability with minimal yearly irradiance variance. System design must account for monsoon precipitation spikes (June–Sept) and high summer ambient temperatures.